

FoodSupply IQ

Distribution Intelligence

Product & Build Overview

African food distribution intelligence – the operational field layer.
A manager web dashboard + an offline-first field-rep mobile app, sharing one FastAPI backend, PostGIS database, and object storage.

Status: Epics 0-4 complete (MVP core). Live on Railway.
Repo: github.com/kofimuad/food_supply_iq
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1. What we're building

FoodSupply IQ is a field-sales + distribution intelligence system for African food distribution (a sister product to AFDP and the DMV African Market Intelligence System). It is the OPERATIONAL field layer: it captures what actually happens on the ground - which stores were visited, which products were sampled, which turned into orders, and which accounts reorder.

There are two apps over one backend:

- Manager web dashboard - the whole market in one place: accounts, the product catalog, the team activity feed, and the conversion funnel.
- Field-rep mobile app - built offline-first for markets and store backrooms with poor signal: log visits, samples and orders, attach photos, and move accounts through the pipeline.

Two roles shape what each person sees:

- MANAGER - sees all accounts and team activity; manages the catalog; views the funnel; edits accounts and contacts.
- REP - scoped to their assigned accounts only; records field work (visits, samples, orders, photos, status changes) from mobile.

2. The core loop: Sample -> Trial -> Repeat

Everything the product does feeds one engine: turning a cold store into a repeat customer. The pipeline a rep moves an account through:

- lead -> in_discussion -> sampled -> trial -> repeat
(or -> not_interested)
- A rep records a SAMPLE given (which products, how many).
- That converts to a TRIAL order (value computed from the catalog).
- Further orders are REPEAT orders; after ≥ 2 orders the account is flagged 'Repeating'.
- The manager funnel shows counts + conversion % at each stage, with drill-down to the accounts sitting in it.

3. Architecture & stack

Backend (api)

- FastAPI + SQLAlchemy (async) + Alembic migrations
- PostgreSQL + PostGIS (geospatial: account/visit locations)
- Redis + arq (background jobs, e.g. nightly repeat-metric rollups)
- JWT auth (access + refresh), argon2 password hashing, manager/rep roles
- MinIO / S3 object storage for visit photos (presigned uploads)

Web (manager dashboard)

- Next.js 14 (App Router) + Tailwind + shadcn-style components
- TanStack Query for data; typed against the API contract

Mobile (field rep)

- Expo / React Native; expo-location (GPS) + expo-image-picker (photos)
- Secure token storage; simple screen navigation

Shared & infra

- @fsiq/api-types - TypeScript types generated from the API's OpenAPI schema, used by both web and mobile (one source of truth)

- Deployed on Railway: api, web, Postgres+PostGIS, Redis, MinIO - auto-deploys on push to main

4. What's been built (Epics 0-4)

Epic 0 - Foundation

- Monorepo (api / web / mobile / shared types); local docker infra
- Auth: login, refresh, manager/rep roles + scoping
- Design system; CI-less auto-deploy pipeline on Railway

Epic 1 - Accounts & Contacts

- Account CRUD with categories, address geocoding (Mapbox), notes
- Account profile: overview / history / contacts tabs
- Contacts: add/edit/remove, one primary, tap-to-call
- Pipeline status transitions with a full change history

Epic 2 - Product Catalog

- Products with pack size + currency-aware price
- Soft-delete (deactivate) and reactivate; search

Epic 3 - Field Visits

- Check-in / log a visit: GPS, outcome tag, notes
- Manager activity feed: visits+samples+orders, rep filter, auto-refresh
- Visit photos: capture on mobile, thumbnails on the web profile

Epic 4 - Samples & Orders (core loop)

- Record samples (product picker + quantities)
- Trial orders (value computed from catalog, linked to the sample)
- Repeat orders + 'Repeating' flag + nightly metric recompute
- Sample -> Trial -> Repeat funnel with conversion % and drill-down

Quality: 37 backend tests, typechecked web + mobile, every story committed; PROGRESS.md tracks status story-by-story.

5. How to navigate - Web (manager)

Sign in at /login. The dashboard is the hub; everything else links from it.

/login	Sign in.
/dashboard	Home. Nav buttons (Accounts, Product catalog, Funnel) + the live Team Activity feed (managers).
/accounts	All accounts. Filter by category / status / name search; add / edit / delete; 'Repeating' badge.
/accounts/:id	Account profile. Tabs:
- Overview	- status control (change pipeline stage), counts, address/location/notes
- History	- visits (outcome, date, notes) + photo thumbnails
- Contacts	- add/edit/remove, mark primary, click to call
/products	Product catalog. Add/edit, deactivate/reactivate, search, show inactive.
/funnel	Sample->Trial->Repeat funnel. Filter by rep + date; click a stage to list its accounts.

6. How to navigate - Mobile (field rep)

The rep app is task-focused around the account a rep is standing in front of.

Login	Sign in as a rep.
My accounts	List of the rep's assigned accounts. Tap one to open it.
Account screen	Everything the rep does in the field: - Stats - visits / samples / orders so far - Change status - tap a stage chip to advance the account - Log a visit - pick an outcome, add notes, 'Check in with GPS' - + Photo - attach a photo to a visit (camera or library) - Record sample - pick products + quantities - Log order - Trial or Repeat, with a live total - Contacts - tap a number to call

7. Live access & credentials

Web	https://web-production-a328bf.up.railway.app
API	https://api-production-19cc.up.railway.app
API docs	<api>/docs (interactive Swagger UI)

Demo logins (password for all: password123):

- manager@foodsupplyiq.com - manager (full access)
- rep.dmv@foodsupplyiq.com - rep (DMV accounts)
- rep.accra@foodsupplyiq.com - rep (Accra accounts)

Note: photos are uploaded from the mobile app and viewed as thumbnails on the web profile. UI polish is a planned later pass.

8. What's next

Remaining MVP epics

- Epic 5 - Map & Territory: rep account map; manager clustering/coverage (PostGIS)
- Epic 6 - Dashboard & KPIs: KPI cards vs target, trends
- Epic 7 - Rep mobile offline-first: today's schedule + offline sync engine

Later (intelligence layer)

- Epic 8 Intelligence, Epic 9 Supply Chain, Epic 10 Advanced, Epic 11 Market Intelligence